```
!pip install librosa seaborn scikit-learn
```

```
Requirement already satisfied: librosa in /usr/local/lib/python3.12/dist-packages (0.11.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: audioread>=2.1.9 in /usr/local/lib/python3.12/dist-packages (from librosa) (3.1
Requirement already satisfied: numba>=0.51.0 in /usr/local/lib/python3.12/dist-packages (from librosa) (0.60.0
Requirement already satisfied: numpy>=1.22.3 in /usr/local/lib/python3.12/dist-packages (from librosa) (2.0.2)
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Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.12/dist-packages (from librosa) (1.5.3)
Requirement already satisfied: decorator>=4.3.0 in /usr/local/lib/python3.12/dist-packages (from librosa) (4.4.2)
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Requirement already satisfied: soxr>=0.3.2 in /usr/local/lib/python3.12/dist-packages (from librosa) (1.0.0)
Requirement already satisfied: typing_extensions>=4.1.1 in /usr/local/lib/python3.12/dist-packages (from librosa) (4.15.0)
Requirement already satisfied: lazy_loader>=0.1 in /usr/local/lib/python3.12/dist-packages (from librosa) (0.5)
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Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /usr/local/lib/python3.12/dist-packages (from seaborn) (3.10.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.0)
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Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.51.0->librosa) (0.43.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.3)
Requirement already satisfied: platformdirs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from pooch>=1.1->librosa) (4.9.4)
Requirement already satisfied: requests>=2.19.0 in /usr/local/lib/python3.12/dist-packages (from pooch>=1.1->librosa) (2.32.4)
Requirement already satisfied: cffi>=1.0 in /usr/local/lib/python3.12/dist-packages (from soundfile>=0.12.1->librosa) (2.0.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.0->soundfile>=0.12.1->librosa) (3.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->pooch>=1.1->librosa) (3
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->pooch>=1.1->librosa) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->pooch>=1.1->librosa) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->pooch>=1.1->librosa) (2026.2.
```

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OK No thanks

```
!wget https://zenodo.org/records/3384388/files/6_dB_valve.zip
!unzip 6_dB_valve.zip
```

```
inflating: valve/id_06/normal/00000940.wav
inflating: valve/id_06/normal/00000941.wav
inflating: valve/id_06/normal/00000942.wav
inflating: valve/id_06/normal/00000943.wav
inflating: valve/id_06/normal/00000944.wav
inflating: valve/id_06/normal/00000945.wav
inflating: valve/id_06/normal/00000946.wav
inflating: valve/id_06/normal/00000947.wav
inflating: valve/id_06/normal/00000948.wav
inflating: valve/id_06/normal/00000949.wav
inflating: valve/id_06/normal/00000950.wav
inflating: valve/id_06/normal/00000951.wav
inflating: valve/id_06/normal/00000952.wav
inflating: valve/id_06/normal/00000953.wav
inflating: valve/id_06/normal/00000954.wav
inflating: valve/id_06/normal/00000955.wav
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inflating: valve/id_06/normal/00000959.wav
inflating: valve/id_06/normal/00000960.wav
inflating: valve/id_06/normal/00000961.wav
inflating: valve/id_06/normal/00000962.wav
inflating: valve/id_06/normal/00000963.wav
inflating: valve/id_06/normal/00000964.wav
inflating: valve/id_06/normal/00000965.wav
inflating: valve/id_06/normal/00000966.wav
inflating: valve/id_06/normal/00000967.wav
inflating: valve/id_06/normal/00000968.wav
inflating: valve/id_06/normal/00000969.wav
inflating: valve/id_06/normal/00000970.wav
inflating: valve/id_06/normal/00000971.wav
inflating: valve/id_06/normal/00000972.wav
inflating: valve/id_06/normal/00000973.wav
inflating: valve/id_06/normal/00000974.wav
inflating: valve/id_06/normal/00000975.wav
inflating: valve/id_06/normal/00000976.wav
inflating: valve/id_06/normal/00000977.wav
inflating: valve/id_06/normal/00000978.wav
inflating: valve/id_06/normal/00000979.wav
inflating: valve/id_06/normal/00000980.wav
inflating: valve/id_06/normal/00000981.wav
inflating: valve/id_06/normal/00000982.wav
inflating: valve/id_06/normal/00000983.wav
inflating: valve/id_06/normal/00000984.wav
inflating: valve/id_06/normal/00000985.wav
inflating: valve/id_06/normal/00000986.wav
inflating: valve/id_06/normal/00000987.wav
inflating: valve/id_06/normal/00000988.wav
inflating: valve/id_06/normal/00000989.wav
inflating: valve/id_06/normal/00000990.wav
inflating: valve/id_06/normal/00000991.wav
```

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```
import os

print(os.listdir())
```

```
['.config', '6_dB_valve.zip', 'valve', 'sample_data']
```

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```
import os
import librosa
import librosa.display
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

```
DATA_PATH = "/content/valve"
```

```
import os
import librosa

DATA_PATH = "/content/valve"

signals = []
labels = []

for root, dirs, files in os.walk(DATA_PATH):
    for file in files:
        if file.endswith(".wav"):   # only audio files
            file_path = os.path.join(root, file)

            try:
                signal, sr = librosa.load(file_path, sr=None)
                signals.append(signal)

                # label = folder name (important)
                label = root.split("/")[-1]
                labels.append(label)

            except Exception as e:
                print("Error loading:", file_path)

print("Total samples:", len(signals))
print("Classes:", set(labels))
```

```
Total samples: 4170
Classes: {'normal', 'abnormal'}
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
plt.plot(signals[0])
plt.title("Waveform (Normal/Abnormal)")
plt.show()
```

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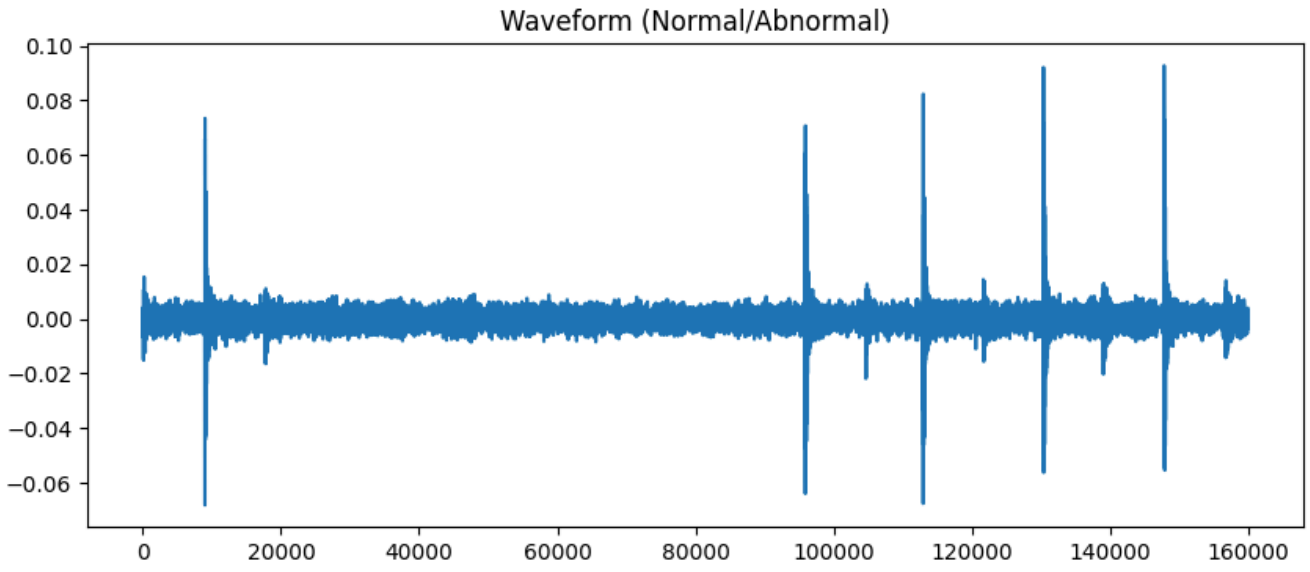
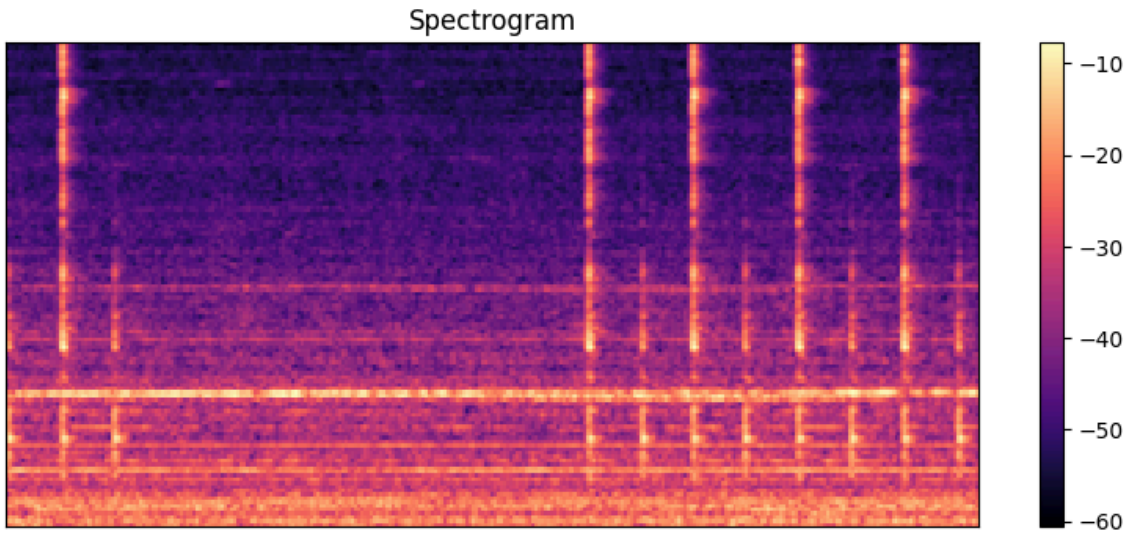



```
import librosa.display

S = librosa.feature.melspectrogram(y=signals[0], sr=sr)

plt.figure(figsize=(10,4))
librosa.display.specshow(librosa.power_to_db(S), sr=sr)
plt.colorbar()
plt.title("Spectrogram")
plt.show()
```



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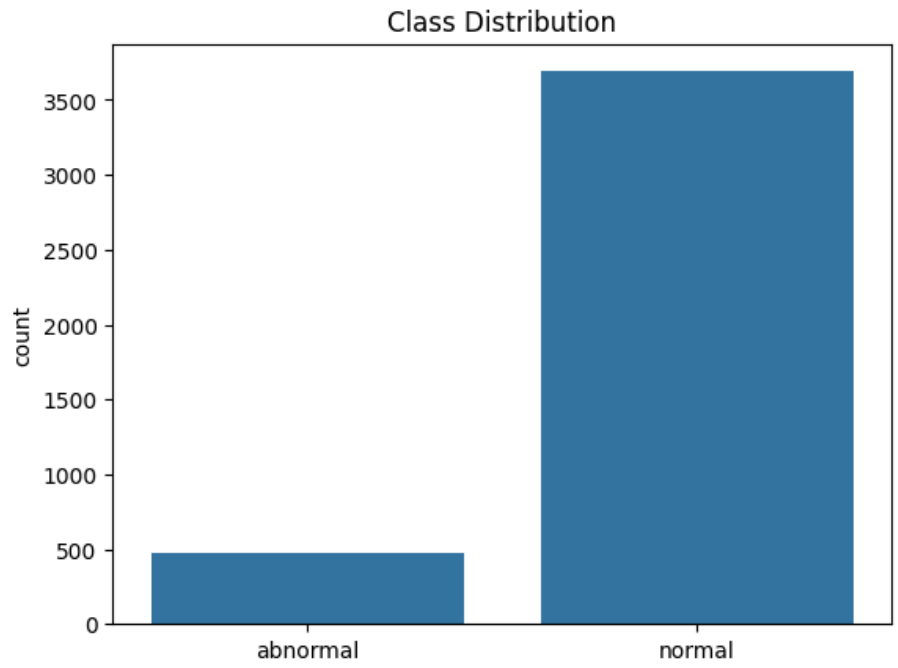
```
import seaborn as sns

sns.countplot(x=labels)
plt.title("Class Distribution")
plt.show()
```



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```
import numpy as np

def extract_features(signal, sr):
    mfcc = librosa.feature.mfcc(y=signal, sr=sr, n_mfcc=13)
    return np.mean(mfcc.T, axis=0)

features = []

for signal in signals:
    features.append(extract_features(signal, sr))

features = np.array(features)

print(features.shape)
```

```
(4170, 13)
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    features, labels, test_size=0.2, random_state=42
)
```

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()
model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier()
```

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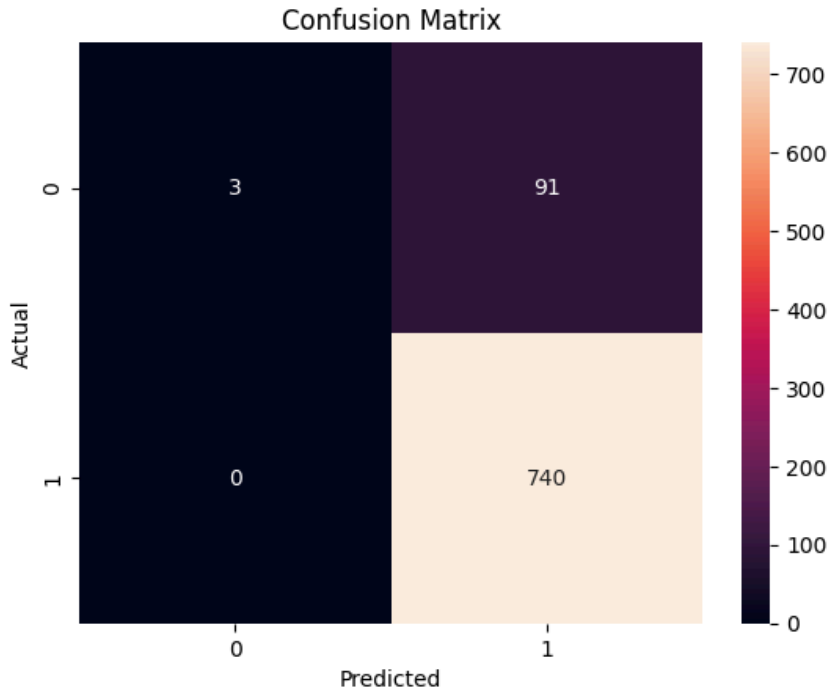
```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.8908872901678657
              precision    recall  f1-score   support

    abnormal       1.00      0.03      0.06        94
      normal       0.89      1.00      0.94       740

    accuracy                           0.89       834
   macro avg       0.95      0.52      0.50       834
weighted avg       0.90      0.89      0.84       834
```

```
import seaborn as sns

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



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```
def add_noise(signal):
    noise = np.random.randn(len(signal))
    return signal + 0.005 * noise

aug_signals = [add_noise(s) for s in signals]
```

```
aug_features = []

for signal in aug_signals:
    aug_features.append(extract_features(signal, sr))

aug_features = np.array(aug_features)

X_train2, X_test2, y_train2, y_test2 = train_test_split(
    aug_features, labels, test_size=0.2
)

model.fit(X_train2, y_train2)

y_pred2 = model.predict(X_test2)

print("New Accuracy:", accuracy_score(y_test2, y_pred2))
```

```
New Accuracy: 0.9088729016786571
```

```
lengths = [len(s) for s in signals]

sns.histplot(lengths, bins=30)
plt.title("Audio Length Distribution")
plt.show()
```

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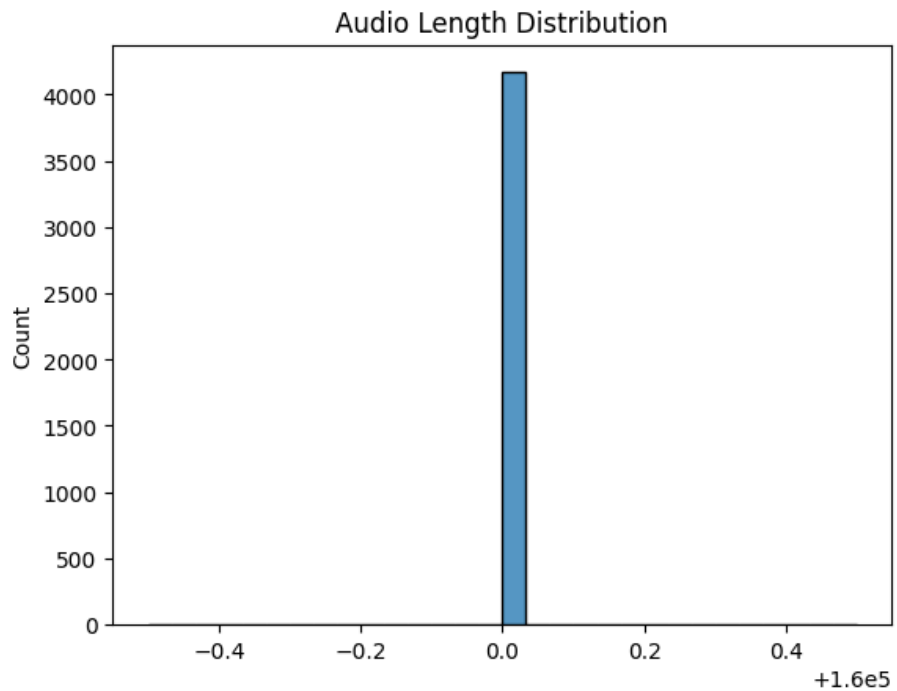
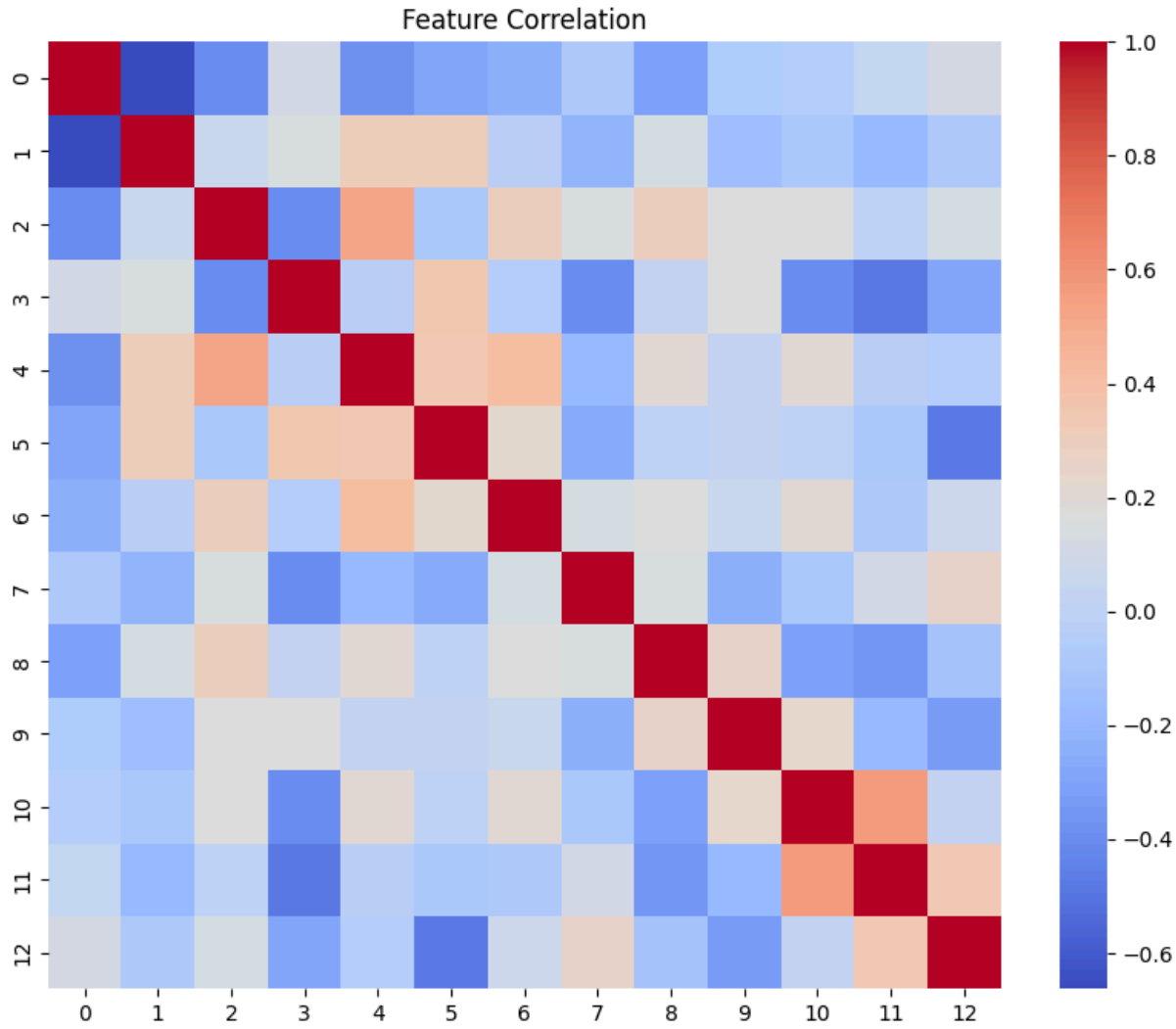



```
import pandas as pd

df = pd.DataFrame(features)

plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), cmap="coolwarm")
plt.title("Feature Correlation")
plt.show()
```



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```
!pip install tensorflow
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from tensorflow) (26.0)
Requirement already satisfied: protobuf!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-
```

```
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.2.0)
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Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.1
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow
Requirement already satisfied: tensorboard~=2.19.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.13
Requirement already satisfied: numpy<2.2.0,>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflo
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.16.
Requirement already satisfied: ml-dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.5.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow) (0.46.3)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow) (0.19.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2026.2.25)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.19.0->tensorflow) (3.10.2)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.19.0->tensorf
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.19.0->tensorflow) (3.1.7)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from werkzeug>=1.0.1->tensorboard~=2.19.0->tensorflow
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow)
```

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```
import os
import numpy as np
import librosa
import librosa.display
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from tensorflow.keras import layers, models
```

```
DATA_PATH = "/content/valve"

X = []
y = []

for root, dirs, files in os.walk(DATA_PATH):
    for file in files:
        if file.endswith(".wav"):
            file_path = os.path.join(root, file)

            signal, sr = librosa.load(file_path, sr=None)

            # Create Mel Spectrogram
            spec = librosa.feature.melspectrogram(y=signal, sr=sr)
```

```
            spec_db = librosa.power_to_db(spec)

            # Resize to fixed shape
            spec_db = np.resize(spec_db, (128, 128))

            X.append(spec_db)

            label = root.split("/")[-1]
            y.append(0 if label == "normal" else 1)

X = np.array(X)
y = np.array(y)

print("Shape:", X.shape)
```

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```
Shape: (4170, 128, 128)
```

```
X = X / np.max(X)

# Add channel dimension for CNN
X = X.reshape(X.shape[0], 128, 128, 1)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = models.Sequential()

# Conv Block 1
model.add(layers.Conv2D(32, (3,3), activation='relu', input_shape=(128,128,1)))
model.add(layers.MaxPooling2D((2,2)))

# Conv Block 2
model.add(layers.Conv2D(64, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))

# Conv Block 3
model.add(layers.Conv2D(128, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))

# Flatten
model.add(layers.Flatten())

# Dense Layers
model.add(layers.Dense(128, activation='relu'))
model.add(layers.Dropout(0.3))
```

```
model.add(layers.Dense(1, activation='sigmoid'))

model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not p
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	320
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 1)	129

Total params: 3,304,193 (12.60 MB)
Trainable params: 3,304,193 (12.60 MB)
Non-trainable params: 0 (0.00 B)

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```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
history = model.fit(
    X_train, y_train,
    epochs=5,
    validation_data=(X_test, y_test)
)
```

```
Epoch 1/5
105/105 ━━━━━━━━━━━━━━━━━━━━ 79s 757ms/step - accuracy: 0.9913 - loss: 0.0263 - val_accuracy: 0.9844 - val_loss: 0.0436
Epoch 2/5
105/105 ━━━━━━━━━━━━━━━━━━━━ 67s 643ms/step - accuracy: 0.9958 - loss: 0.0180 - val_accuracy: 0.9916 - val_loss: 0.0188
```

```
Epoch 3/5
105/105 ━━━━━━━━━━━━━━━━━━━━ 70s 671ms/step - accuracy: 0.9970 - loss: 0.0093 - val_accuracy: 0.9928 - val_loss: 0.0188
Epoch 4/5
105/105 ━━━━━━━━━━━━━━━━━━━━ 84s 690ms/step - accuracy: 0.9985 - loss: 0.0073 - val_accuracy: 0.9916 - val_loss: 0.0221
Epoch 5/5
105/105 ━━━━━━━━━━━━━━━━━━━━ 82s 780ms/step - accuracy: 0.9991 - loss: 0.0061 - val_accuracy: 0.9916 - val_los
```

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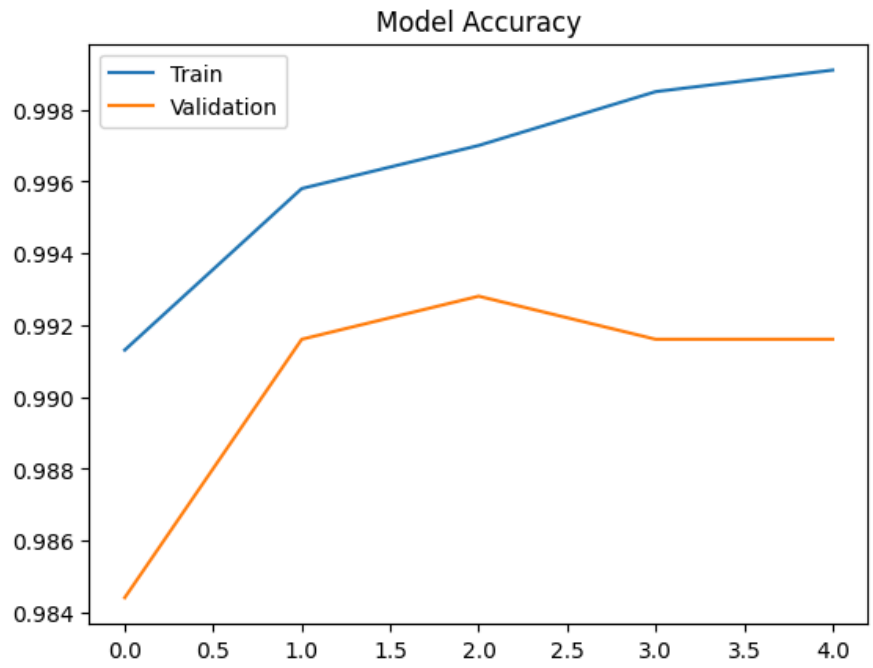
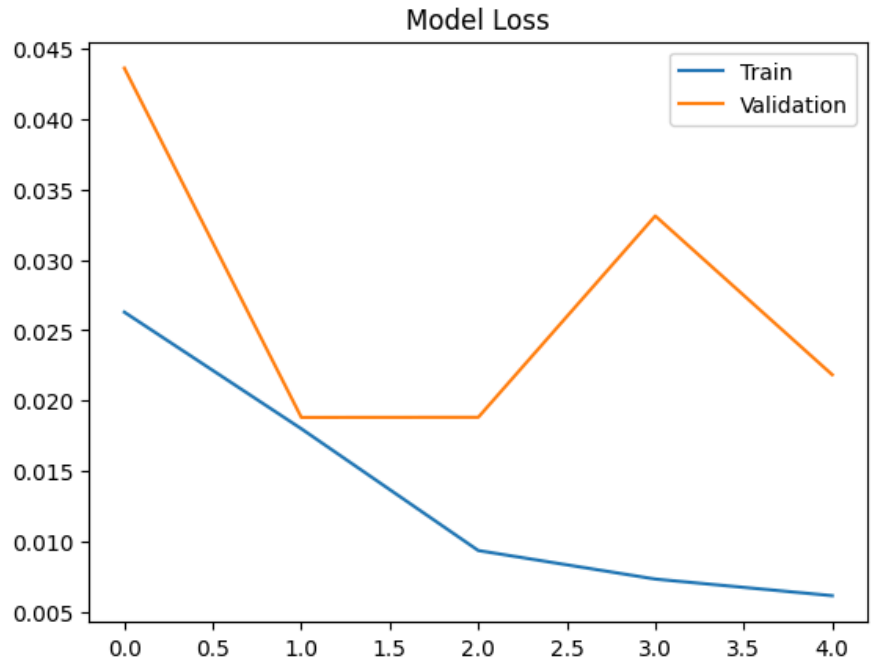
```
loss, acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", acc)
```

```
27/27 ━━━━━━━━━━━━━━━━━━━━ 8s 315ms/step - accuracy: 0.9916 - loss: 0.0218
Test Accuracy: 0.9916067123413086
```

```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title("Model Accuracy")
plt.legend(["Train", "Validation"])
plt.show()
```




```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title("Model Loss")
plt.legend(["Train", "Validation"])
plt.show()
```



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